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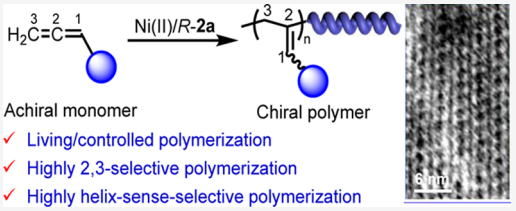
Highly Regioselective and Helix-Sense Selective Living Polymerization of Phenyl and Alkoxyallene Using Chiral Nickel(II) Catalysts

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ABSTRACT: Facile synthesis of high optically active helical polymers from achiral monomers still remains a great challenge. In this work, two chiral Ni(II) catalysts bearing chiral *R*- or *S*-**2a** ligands were facilely prepared which can efficiently initiate the living/controlled polymerization of allene monomers and afford well-defined polyallenes in high yields with controlled molecular masses (M_ns) and low molecular mass distributions (M_w/M_ns). The catalysts showed interesting regioselectivity on the cumulated double bonds of allene monomers, as the polymerizations were highly selective which occurred on 2,3-double bonds. Moreover, the polymerization of achiral allene monomers using chiral Ni(II) catalysts showed remarkable helix-sense selectivity. The yielded polyallenes exhibited high optical activity due to the formation of a predominant one-handed helix as revealed by circular dichroism, UV−vis, and direct atomic force microscopy (AFM) observations as well. The predominant one-handed helicity of the generated polyallenes was solely determined by the chirality of the Ni(II) catalysts.



INTRODUCTION

Helix is one of the most important secondary structural motifs in biopolymers and also in synthetic macromolecules.[1−3] The one-handed helical conformation adopted by biological macromolecules such as protein and DNA plays important living functions including molecular recognition, catalysis, inheritance, replication, and so on.[4−6] A slight damage on the helical structure of biomacromolecules can cause a variety of severe diseases. For example, misfolding the helix of proteins into an extended β-sheet may result in amyloid structures accompanied by grievous dysfunctions including Alzheimer's disease, Parkinson's disease, and diabetes.[7,8] Therefore, stimulated by the biological helices and functions, synthetic helical polymers have attracted great attention from synthetic chemistry to materials science and biological engineering.[9−13] These research studies not only explore the structure and property of the unique helices in nature but also develop new materials with novel functions.[14−17] Moreover, the study on a helical polymer can bring out a new structural platform for the development of novel functional materials.[18−21] Although the helical polymer is still on the preliminary research stage, it has exhibited a wide range of applications, such as those in enantiomer separation, chiral switch, asymmetric catalysis, circularly polarized luminescence (CPL), and optoelectronic devices among others.[22−26] From the reported helical polymers, polyallene is of particular interest because it contains reactive *exo*-double bonds along the backbone which can be further decorated with other intriguing functionalities.[27,28] Polyallene is generally produced through the radical polymerization or coordination polymerization of the corresponding allene monomers using transition-metal complexes as catalysts.[29,30] The living/controlled coordination polymerization can yield polyallenes with expected molecular mass and low polydispersity.[31,32] However, both the accumulated double bonds of allene monomers can be polymerized and result in polyallene with complex architectures.[33−36] Thus, developing a new method for living/controlled polymerization of allene monomers with high regioselectivity of the accumulated double bonds is greatly desired.

On the other hand, optically active helical polyallenes were usually prepared from the polymerization of chiral allene monomers with high enantiomeric purity.[37] The helix-sense was determined by the chirality of the monomers, and a large amount of chiral materials was consumed during the polymerization.[38] As we know, enantiopure materials are generally limited and expensive except for those derived from the natural product, which may further limit the type and scope of the available monomers. Thus, synthesis of large optically active helical polyallene from achiral materials is greatly intriguing because a plenty of optically active polymers

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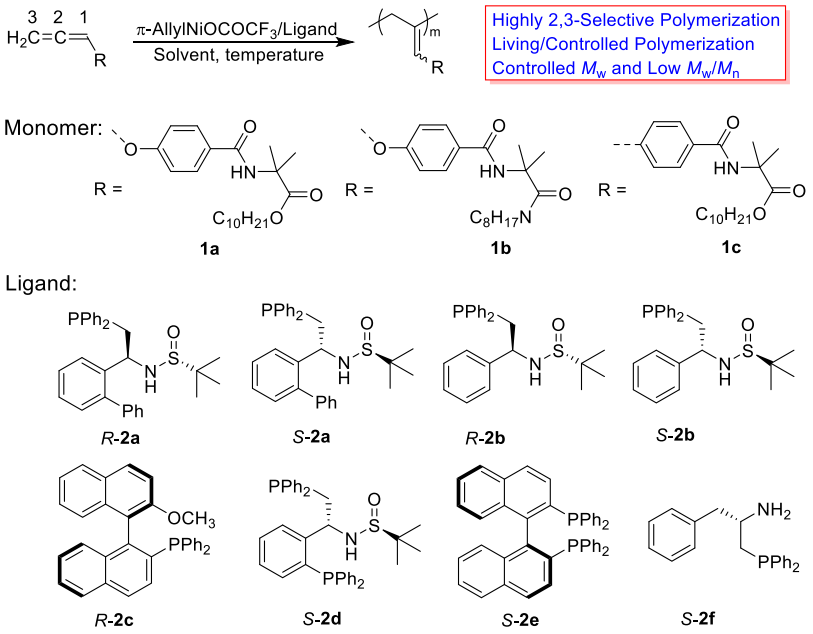


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Scheme 1. Highly Regioselective and Helix-Sense Selective Living Polymerization of Allene Monomers



can be facilely obtained from a tiny amount of chiral catalysts, initiators, or additives through chirality transfer and amplification strategies.[39] Moreover, such kind of helical polymers can exhibit high optical activity derived solely from one-handed helicity, without containing any other chiral moieties on the polymer pendants or on the chain ends. However, helical polyallene prepared from achiral allene monomers has seldom been reported owing to the limited polymerization strategy albeit this strategy possessing many advantages. Thus, exploring a novel catalyst or initiator for controlled synthesis of optically active helical polymers from a achiral allene monomer with high region-selectivity is very interesting.

Herein, we report on a family of chiral nickel(II) catalysts which can initiate the living polymerization of various allene monomers with high region and helix-sense selectivity. The polymerization occurred with high selectivity on the 2,3-double bond among the two accumulated double bonds of allenes. Moreover, the polymerization of achiral allene monomers by the chiral Ni(II) catalysts produced predominantly one-handed helical polyallene with large optical activity due to the asymmetric induction of the chiral catalysts.

RESULTS AND DISCUSSION

First, to screen out an excellent chiral Ni(II) catalyst, a family of chiral phosphine ligands (**2a**−**2f**) were designed and synthesized (Scheme 1). After full characterization, these ligands were treated with (π-allyl)NiOCOCF$_3$ in toluene at room temperature to produce the desired Ni(II) catalysts.[40,41] The ratio of the ligand to Ni(II) was tentatively fixed at 1/1. The resulting catalyst solutions were directly used in the polymerization of allene monomers without further isolation and purification steps to lower the risk of catalyst decomposition. Three allene monomers **1a**−**1c** were designed and synthesized followed the literature with some modifications (see Supporting Information for the details of synthesis).[37] The activity of the Ni(II) catalysts was investigated through the polymerization of allene monomer **1a** in CH_2Cl_2 at 25 °C ($[\mathbf{1a}]_0/[Ni]_0$ = 100, $[\mathbf{1a}]_0$ = 0.2 M). After most of the monomer was assumed, the polymerization solution was precipitated into methanol and the yielded polymers were collected via filtration. The isolated polymers were first analyzed by size exclusion chromatography (SEC), and the results were summarized in Table 1. It was found that all the

Table 1. Polymerization Results for Monomer 1a Initiated by Different (π-allyl)NiOCOCF$_3$/Ligand Catalysts[a]

run	ligand	T (°C)	Time (h)[b]	M_n[c] (kDa)	M_w/M_n[c]	yield[d] (%)
1	R-2a	25	15	40.2	1.05	90
2	R-2b	25	25	40.5	1.06	85
3	R-2c	35	96	40.3	1.24	76
4	S-2d	25	18	40.1	1.05	74
5	S-2e	35	96	39.9	1.27	72
6	S-2f	35	50	40.4	1.15	81

[a]The polymerizations were carried in CH_2Cl_2 with $[\mathbf{1a}]_0$ = 0.2 M, and $[\mathbf{1a}]_0/[Ni]_0$ = 100. [b]The time was the consumption of the monomer almost completely. [c]The M_n and M_w/M_n were determined by SEC. [d]The isolated yields.

Ni(II) catalysts can promote the polymerization of monomer **1a**, and polymers with similar molecular mass (M_n = ca. 40 kDa) and low polydispersity (M_w/M_n < 1.28) were afforded. However, the catalysts with different ligands also showed different behaviors on the polymerizations. The ligand *R*-**2a** showed highest activity among others because it took a short polymerization time (ca. 15 h) and gave the expected polymer in high yield with expected M_n and quite low dispersity with M_w/M_n = 1.05 (run 1, Table 1). The Ni(II) catalysts bearing *R*-**2c** and *S*-**2e** ligands showed a lower activity as the polymerization took more than three days even at higher temperatures, and the yields were low (run 3 and run 5 in Table 1; Figure S18 in Supporting Information). Thus, the Ni(II) catalyst bearing the *R*-**2a** ligand was the good catalyst for the polymerization of allene monomer **1a**. The influence of the ratio of the *R*-**2a** ligand to Ni(II) was also investigated. It was found that the best ratio of *R*-**2a** to Ni(II) was 1/1; an increase or decrease of the ratio would broaden the molecular

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mass distribution (Figure S19 and Table S1 in Supporting Information).

The behaviors of the polymerizations initiated by Ni(II)/*R*-**2a** were studied in more detail. A couple of polymerizations of **1a** initiated by Ni(II)/*R*-**2a** were parallelly carried out in CH_2Cl_2 at room temperature but under different molar ratios of the monomer to the Ni(II) catalyst ($[\mathbf{1a}]_0/[Ni]_0$). All the polymerizations afforded the expected polymers in high yields (>85%) with unimodal and symmetrical SEC traces (Figure 1a). The elution peaks moved to a low retention-time region

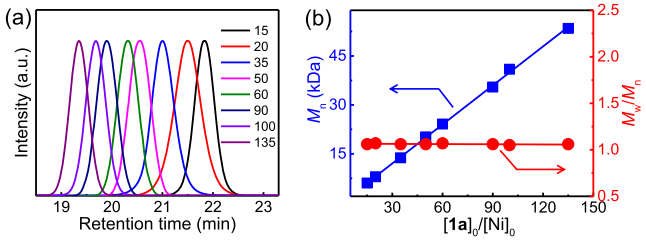


Figure 1. (a) SEC traces of *R*-poly-$\mathbf{1a}_m$s synthesized through the polymerization of **1a** in different feed ratios of **1a** to Ni(II)/*R*-**2a** catalyst. (b) Plots of M_n and M_w/M_n values against the value of $[\mathbf{1a}]_0/[Ni]_0$.

with an increased $[\mathbf{1a}]_0/[Ni]_0$ ratio. The M_n of the generated polymers showed a linear correlation to the initial ratio of $[\mathbf{1a}]_0/[Ni]_0$, and all the isolated polymers had low dispersity with $M_w/M_n < 1.08$ (Figure 1b; run 1−8 in Table 2). This result suggests that the polymerization of **1a** initiated by Ni(II)/*R*-**2a** proceeded in a chain-growth living/controlled fashion. SEC tracking experiment of the polymerization further supported the living fashion of the polymerization ($[\mathbf{1a}]_0 = 0.2$ M, $[\mathbf{1a}]_0/[Ni]_0 = 50$). As displayed in Figure 2, the SEC peak of the generated polyallene continually moved to a higher M_n-area and remained monodisperse. The plot of the M_n value against the monomer conversion revealed that the M_n increased linearly to the monomer conversion while the M_w/M_n kept narrow throughout the whole polymerization process (Figure 3a). Interestingly, the polymerization was quite fast; >90% of monomer was consumed within 8 h. Kinetic analysis demonstrated that the polymerization agreed with the first-

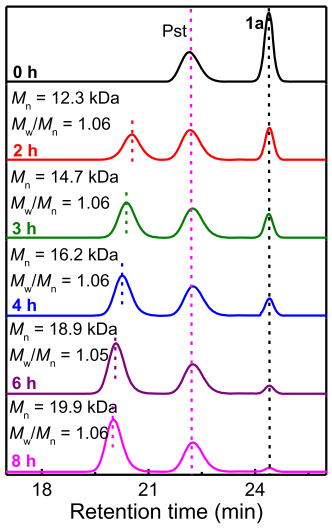


Figure 2. Time-dependent SEC for the polymerization of **1a** initiated by Ni(II)/*R*-**2a** with polystyrene (Pst) as the internal standard in CH_2Cl_2 at 25 °C ($[\mathbf{1a}]_0 = 0.2$ M, $[\mathbf{1a}]_0/[Ni]_0 = 50$).

order reaction mechanism, and the polymerization rate constant was 7.8×10^{-5} s^{-1} (Figure 3b). These results clearly indicated that the Ni(II) catalyst bearing a chiral phosphine ligand *R*-**2a** was an excellent catalyst for the living polymerization of allene monomer **1a**. Thus, this initiating system of Ni(II)/*R*-**2a** was further extended to the polymerization of monomer **1b** and **1c** under similar experimental conditions to those of **1a**. As expected, the polymerizations proceeded in living/controlled fashions and gave desired polyallenes in high yields with tunable M_n and relatively low M_w/M_n (run 11−14 in Table 2; Figure S21−S22 and Table S2 in Supporting Information). These studies further verified the high catalytic activity of the Ni(II)/*R*-**2a** catalyst.

The chemical structure of the generated polymers was investigated by ^1H, ^{31}P, and ^{13}C NMR and FT-IR as well. ^1H NMR spectra of monomer **1a** and the resulting *R*-poly-$\mathbf{1a}_{100}$ recorded in $CDCl_3$ are displayed in Figure 4a,b, respectively. In contrast to the sharp signals of **1a**, the resonances of the resulting polymer were quite broad. The protons of the

Table 2. Polymerization of 1a−1c Using Ni(II)-Catalysts Bearing *R*- and *S*-2a Ligands

run	monomer	ligand[a]	polymer[b]	M_n[c] (kDa)	M_w/M_n[c]	yield[d] (%)	g_{265}[e] ($\times 10^{-4}$)	$[\alpha]_{25}^{D}$[f]
1	**1a**	*R*-**2a**	*R*-poly-$\mathbf{1a}_{15}$	6.1	1.06	70	−0.97	−7.7
2	**1a**	*R*-**2a**	*R*-poly-$\mathbf{1a}_{20}$	8.0	1.07	73	−1.21	−9.4
3	**1a**	*R*-**2a**	*R*-poly-$\mathbf{1a}_{35}$	13.8	1.06	85	−1.67	−13.2
4	**1a**	*R*-**2a**	*R*-poly-$\mathbf{1a}_{50}$	20.2	1.06	85	−2.12	−17.1
5	**1a**	*R*-**2a**	*R*-poly-$\mathbf{1a}_{60}$	24.1	1.07	87	−2.30	−19.4
6	**1a**	*R*-**2a**	*R*-poly-$\mathbf{1a}_{90}$	35.5	1.06	88	−2.61	−22.4
7	**1a**	*R*-**2a**	*R*-poly-$\mathbf{1a}_{100}$	40.2	1.05	90	−2.67	−22.9
8	**1a**	*R*-**2a**	*R*-poly-$\mathbf{1a}_{135}$	53.4	1.06	90	−2.67	−23.0
9	**1a**	*S*-**2a**	*S*-poly-$\mathbf{1a}_{100}$	40.5	1.06	90	+2.64	+22.9
10	**1a**	*S*-**2a**	*S*-poly-$\mathbf{1a}_{135}$	52.9	1.06	89	+2.70	+23.0
11	**1b**	*R*-**2a**	*R*-poly-$\mathbf{1b}_{100}$	39.6	1.07	84	−2.46	−20.1
12	**1b**	*S*-**2a**	*S*-poly-$\mathbf{1b}_{100}$	39.4	1.08	82	+2.49	+20.2
13	**1c**	*R*-**2a**	*R*-poly-$\mathbf{1c}_{100}$	38.9	1.08	81	−2.52	−20.4
14	**1c**	*S*-**2a**	*S* -poly-$\mathbf{1c}_{100}$	39.2	1.09	83	+2.55	+20.6

[a]The polymerizations were performed in CH_2Cl_2 at 25 °C. [b]The footnotes indicate the initial feed ratio of the monomer to catalyst. [c]The M_n and M_w/M_n values were determined by SEC using polystyrene standards. [d]The isolated yields. [e]The asymmetric factor at 265 nm (g_{265}) calculated from CD and UV−vis spectra (c = 0.045 mg/mL). [f]The optical rotation was recorded in THF at room temperature (c = 0.5 mg/mL).

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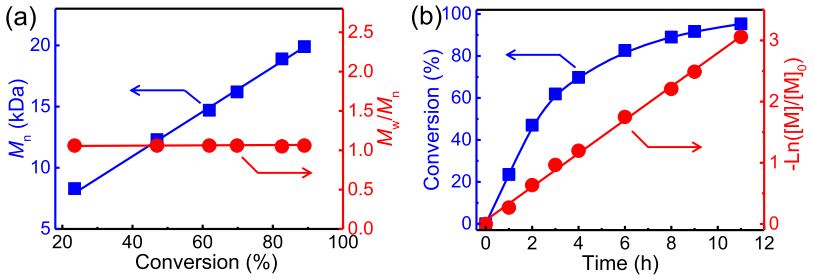


Figure 3. (a) Plots of M_n and M_w/M_n values against the monomer conversion for the polymerization of **1a** initiated by Ni(II)/*R*-**2a** in CH_2Cl_2 at 25 °C ($[\mathbf{1a}]_0$ = 0.2 M, $[\mathbf{1a}]_0/[Ni]_0$ = 50). (b) Plots of the monomer conversion and -Ln($[M]/[M]_0$) values against the polymerization time.

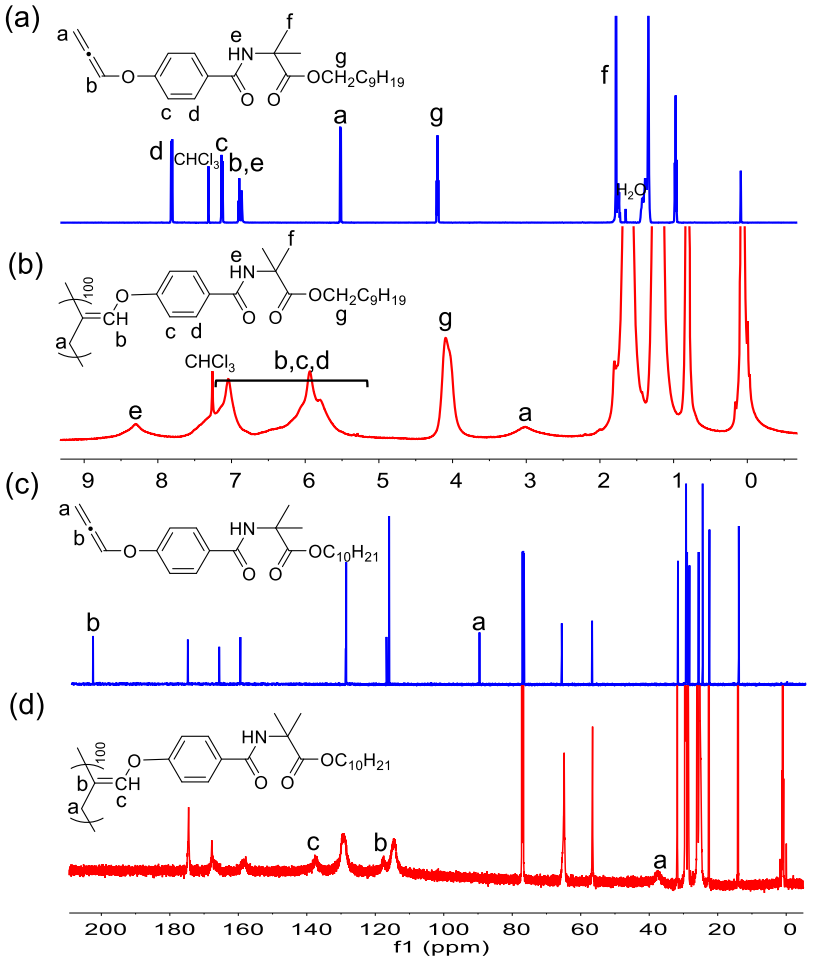


Figure 4. ^{1}H NMR (600 MHz) spectra of **1a** (a) and the resulting *R*-poly-$\mathbf{1a}_{100}$ (b) and ^{13}C NMR (150 MHz) of **1a** (c) and *R*-poly-$\mathbf{1a}_{100}$ (d) recorded in $CDCl_3$ at 25 °C.

terminal double bond (2,3-double bond, *a*) of the monomer completely disappeared, indicative of the full polymerization of the terminal double bonds. Accordingly, a new broad resonance appeared at 3.02 ppm on the ^{1}H NMR spectrum of the *R*-poly-$\mathbf{1a}_{100}$ polymer, derived from the *exo*-proton of the polymerized double bond. Interestingly, no peak can be detected around 5.0 ppm, corresponding to the polymerization of 1,2-double bonds of **1a**.[42,43] This result confirmed that the polymerization of the allene monomer **1a** mainly occurred on the 2,3-double bonds, suggesting the high regioselectivity of the polymerization. The unreacted 1,2-double bond of the monomer was overlapped with the phenyl protons and cannot be clearly discerned. It is worthy to note that the resonance of N−H shifted to the downfield compared to the monomer **1a**. This is reasonable because of the formation of the intramolecular hydrogen bonding between the N−H and C═O of adjacent repeating units, which further supports the generation of expected polymer. The existence of intramolecular hydrogen bonding between the N−H and C═O of adjacent repeating units could also be further confirmed by the FT-IR spectra of *R*-poly-$\mathbf{1a}_{100}$ measured in toluene and $CHCl_3$ solution, respectively (Figure S23 in Supporting Information). ^{13}C NMR analyses further verified the formation of the expected polymer. As displayed in Figure 4d, the resonance of the carbons of *R*-poly-$\mathbf{1a}_{100}$ was quite clear; only the 2,3-polymerized product can be clearly observed, confirming the high 2,3-selective polymerization of the catalyst. In order to figure out the syn or anti configuration of the double bond of

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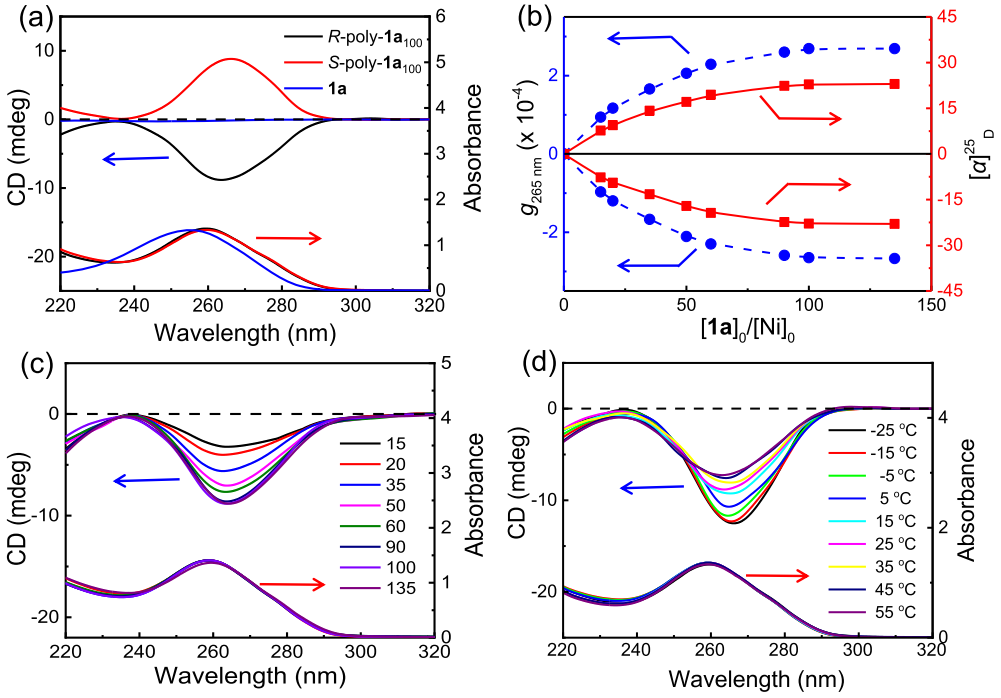


Figure 5. (a) CD and UV−vis spectra of monomer **1a** and *R*- and *S*-poly-**1a**$_{100}$. (b) Plots of g_{265} intensity and optical rotation of *R*- and *S*-poly-**1a**$_m$s with different M_n. The CD and UV−vis spectra were recorded in THF at room temperature with c = 0.045 mg/mL. (c) M_n-dependent CD and UV−vis spectra of *R*-poly-**1a**$_m$. (d) Temperature-dependent CD and UV−vis spectra of *R*-poly-**1a**$_{100}$.

the pendant, a series of NMR and FT-IR spectra were recorded; however, it was hard to determine the syn or anti configuration of the double bond. Furthermore, ^{1}H, ^{31}P NMR, IR, and CD spectra were recorded to confirm whether the Ni(II)-complex resided on the end chain of the isolated polyallenes. However, no detectable signals belonging to the phosphine ligand could be observed, even on the polymers with very low M_n. Particularly, no signals of the ligand on ^{31}P NMR could be discerned, confirming that the Ni(II)-complex was decomposed from the chain end of the polymer during the process of precipitation (Figure S24 in Supporting Information). Probably, the phosphine ligand was oxidized during the postpolymerization workup, and the Ni(II)-complex was decomposed and left from the polymer chain end. The living/controlled polymerizations of monomer **1b** and **1c** by Ni(II)/*R*-**2a** all occurred at the 2,3-double bond as demonstrated by ^{1}H NMR analyses (Figure S11 and Figure S14 in Supporting Information). Collectively, these studies support the high efficiency and regioselectivity of the catalyst.

Since the polymerization gave an expected well-defined polymer with anticipated M_n, narrow M_w/M_n, and high 2,3-selectivity, the chiroptical properties of the generated polymers were further investigated. Although monomer **1a** was achiral and circular dichroism (CD) silent, the isolated *R*-poly-**1a**$_{100}$ showed large optical activity as revealed by CD and UV−vis absorption analyses. The CD and UV−vis spectra of monomer **1a** and *R*-poly-**1a**$_{100}$ are displayed in Figure 5a. Monomer **1a** showed strong absorption with the maximum located at 255 nm. While the resulting *R*-poly-**1a**$_{100}$ showed absorption at a relatively longer wavelength region, the absorption band around 260 nm of *R*-poly-**1a**$_{100}$ may also contribute to the $\pi-\pi^*$ transition of the benzene ring. As anticipated, monomer **1a** was CD silent while the produced *R*-poly-**1a**$_{100}$ exhibited distinct CD at the backbone absorption region, indicative of the predominant one-handed helix adopted by the polymer backbone. The CD intensity increased with the increase of the degree of the polymerization (DP) until the M_n reached to 40.2 kDa, suggesting that a stable helix was formed when the DP reached to ca. 100. The excess of the left-handed helix of *R*-poly-**1a**$_{100}$ was calculated to be ca. 10% compared to the single left-handed polyallene reported by our group previously.[37] This result suggested that a predominated left-handed helix was selectively produced using *R*-**2a** as the ligand in the polymerization. Although the helix-sense excess of the *R*-poly-**1a**$_{100}$ was not high, this was one of the few methods to obtain chiral polyallenes from achiral monomers.

The optical activity of the polymers measured by optical rotation revealed a similar relationship between the optical activity with the M_n (Figure 5b,c). The diluting experiment demonstrated that the CD and absorption intensities of *R*-poly-**1a**$_{100}$ decreased linearly with the diluted concentration, while the g_{265} was constant. These results indicated that the helical structure was maintained during the dilution and also supported that the optical activity comes from the polymer itself and not the intermolecular self-assembly or aggregation (Figure S25 in Supporting Information). To further verify that the optical activity was attributed to the asymmetric induction of the chiral Ni(II) catalyst, *S*-poly-**1a**$_{100}$ was prepared under the same conditions through living and regioselective polymerization of **1a** using Ni(II)/*S*-**2a** as the catalyst, the enantiomer of Ni(II)/*R*-**2a**. For comparison, CD and UV−vis spectra of *S*-poly-**1a**$_{100}$ were recorded and also displayed in Figure 5a. It was found that *S*-poly-**1a**$_{100}$ showed almost the same absorption behavior to that of *R*-poly-**1a**$_{100}$, suggesting that the chemical structures of the two polymers were almost the same. Interestingly, *S*-poly-**1a**$_{100}$ also exhibited an intense CD at the absorption region of the backbone but was mirror imaged to that of *R*-poly-**1a**$_{100}$ due to the opposite handedness of the helical backbone to that of *R*-poly-**1a**$_{100}$. This result verified that the optically active helical backbone of these

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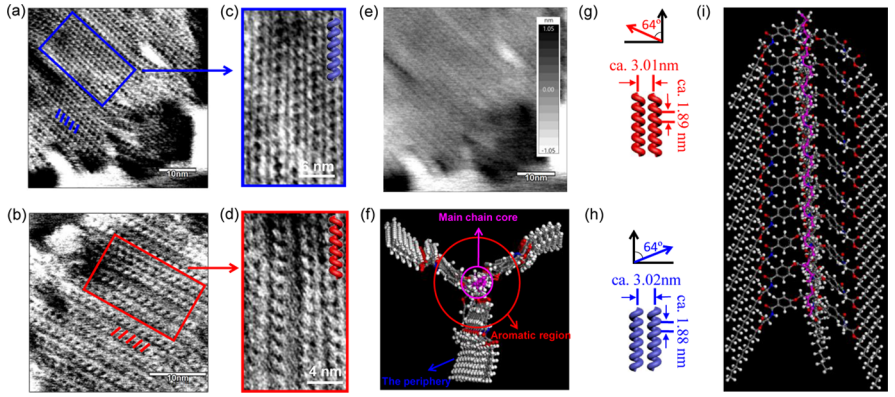


Figure 6. (a) AFM phase images of *S*-poly-**1a**$_{100}$ (a) and *R*-poly-**1a**$_{100}$ (b) and the zoomed images of *S*-poly-**1a**$_{100}$ (b) and *R*-poly-**1a**$_{100}$ (d). (e) AFM height image of *S*-poly-**1a**$_{100}$. (f) Computer-simulated helical conformations of *R*-poly-**1a**$_{100}$ with the top view. Schematic representations of the right-handed helical left-handed helical *R*-poly-**1a**$_{100}$ (g) and *S*-poly-**1a**$_{100}$ (blue) (h). Computer-simulated helical conformations of *R*-poly-**1a**$_{100}$ with the side view (i).

polymers indeed comes from the asymmetric induction of the chiral catalyst during the polymerization. The helical structure of the generated polymers was quite stable and exhibited intense CD in various organic solvents, such as tetrahydrofuran (THF), CH_2Cl_2, $CHCl_3$, and so forth. (Figure S26 in Supporting Information). The CD intensity varied with the polarity of the solvents probably due to the variation on the hydrogen bonding strength between the adjacent repeating units in solvents. Accordingly, the g_{265} value of *R*-poly-**1a**$_{100}$ in THF decreased from 3.10×10^{-4} to 1.81×10^{-4} when the temperature of the solution was increased from −25 to 55 °C, and the g_{265} value of *R*-poly-**1a**$_{100}$ could be increased to original intensity upon the solution cooling to −25 °C. which suggested the dynamic and helix memory characters of the synthesized helical polyallene (Figure 5d and Figure S27 in Supporting Information).[44] Then, the CD intensity of *R*-poly-**1a**$_{100}$ was measured in THF and $CHCl_3$ alternately; it could be observed that the solvent-dependent CD changes were reversible too (Figure S28 in Supporting Information). The slight CD variations on the temperature and solvents may be ascribed to the racemization of the helix-sense at higher temperatures and polar solvents, which also suggested that the polyallene may have partial dynamic and chiral memory properties. However, the precise mechanism of the helix-revision is not very clear under the current stage. The helix-sense selective manner of the living polymerization initiated by the Ni(II)/*R*-**2a** can be extended to monomer **1b** and **1c**. The produced polymers also exhibited intense CD at the backbone absorption region (Figure S29 in Supporting Information) because of the formation of one-handed predominant helices induced by the chiral catalyst. It is worthy to note that the large optical activities of the synthetic helical polyallene solely come from the helicity of the main chain because there were no chiral moieties on the pendants or at the chain ends. The chiral phosphine ligand was removed during the postpolymerization workup process. Therefore, it can be concluded that the chirality economy of the polymerizations using the chiral Ni (II) catalyst was quite high. That is, the catalytic amount of the chiral ligand can produce a large amount of chiral materials with high optical activity.

In order to get more details, helical structures of *S*-poly-**1a**$_{100}$ and *R*-poly-**1a**$_{100}$ were further convinced by atomic force microscope (AFM) investigations. The dilute solutions of *S*-poly-**1a**$_{100}$ and *R*-poly-**1a**$_{100}$ in THF were cast onto highly oriented pyrolytic graphite (HOPG) and slowly evaporated to dryness in air.[45−47] Then, AFM observations were performed, and the recorded images were displayed in Figure 6a,b. As expected, right- and left-handed twisted fibers were observed clearly on the samples of *S*-poly-**1a**$_{100}$ and *R*-poly-**1a**$_{100}$, respectively. Meanwhile, from the zoomed AFM images, the helix-sense of each polymer chain could be clearly distinguished (Figure 6c,d). The average height and diameter of the helical chain of *S*-poly-**1a**$_{100}$ were estimated to be ca. 2.52 and 2.17 nm, respectively (Figure 6a; 6e and Figure S30 in Supporting Information). The space between neighboring polymer chains was ca. 3.02 nm, while the helical pitch was ca. 1.88 nm (Figure 6g,h). Moreover, the helical array of the pendants in clockwise direction can be clearly observed at 64 ± 2° relative to the backbone axis from the high-resolution AFM images. Accordingly, the diameter and helical pitch of *R*-poly-**1a**$_{100}$ were ca. 2.10 and 1.89 nm, respectively, while the helical array of the pendants was anticlockwise with 64 ± 2°, similar to those of *S*-poly-**1a**$_{100}$, suggesting the formation of stereo-regular helical polyallenes (Figure 6g,h).

Beside CD and AFM analyses, computer simulation of *R*-poly-**1a**$_{100}$ was carried out to further confirm the helix of the main chain. As shown in Figure 6f,i, the views of the three-dimensional optimal structure from the top and side are shown using a 30-mer of *R*-poly-**1a**$_m$ as a model. It could be seen that a 3_1 rod-like single-handed helical conformation was formed, which was the most stable conformation of *R*-poly-**1a**$_m$. There were three parts including the aromatic region, the periphery, and the twisting main chain core in the polymer chain of *S*-poly-**1a**$_m$ (Figure 6f). This structure could be considered as columns like the helical polymer with similar structures. The structure could be regarded similar to a helical polymer having a similar structure. At last, due to the rigid rod-like feature, *R*-poly-**1a**$_{100}$ and *S*-poly-**1a**$_{100}$ exhibited a liquid crystal phase as revealed by polarized optical micrographs obtained in the concentrated THF solutions at room temperature (Figure S31 in Supporting Information).

■ CONCLUSIONS

In summary, we have developed two new kinds of chiral Ni(II) catalysts, which can initiate a living/controlled polymerization of various allene monomers and generate expected polymers in high yields with tunable M_ns and low M_w/M_ns. Remarkably, the polymerizations showed very high regioselectivity, and the

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polymerizations mainly occurred on the 2,3-double bond of the allene monomers. Moreover, the living polymerizations of achiral allene monomers by chiral Ni(II) catalysts exhibited an interesting helix-sense selectivity, generating high optically active helical polyallene with a predominant one-handed helix. The helicity of the polyallene was confirmed by CD, UV−vis, and direct AFM observations. We believe that the present study provides not only a new method for controlled synthesis of optically active helical polymers but also a family of functional materials with great potentials in asymmetric catalysis, CPL, and so forth.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.0c02198.

Synthetic procedure and additional spectroscopies (PDF)

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Notes

The authors declare no competing financial interest.

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